AATIF JUNAID

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B.Tech: Electronics and Communication Engineering
GitHub | linkedin.com/in/aatifjunaid/ | Portfolio | Leetcode

SUMMARY

Final-year B.Tech ECE student targeting processor design and verification. Skilled in RTL design, pipelined processor architecture, constrained-random verification, SVA and functional coverage.

EDUCATION

BTech	2023-2027	Netaji Subhas University of Technology	8.13
Class XII	2023	White Leaf Public School, Delhi	92.6%
Class X	2021	Delhi Public School, Greater Noida	94.6%

PROJECTS

5-Stage Pipelined RISC-V (RV32I) Core | Verilog, SystemVerilog, SVA: GitHub

- Designed a 5-stage pipelined core supporting 8 instructions, with data forwarding, a 1-cycle load-use stall and a 2-instruction branch flush.
- Verified with a self-checking testbench running 5 assembly programs against a single-cycle reference model, comparing all 32 registers and 256 memory words (5/5 passing).
- Added 4 SVA assertions (x0 = 0, stall only on load-use, flush only on taken branch, bubble insertion).

Synchronous FIFO Verification | SystemVerilog, SVA, Functional Coverage, Cadence Xcelium: GitHub

- Developed a class-based, self-checking testbench (constrained-random generator, driver, monitor, queue-based scoreboard) for a parameterized synchronous FIFO.
- Implemented 17 bound SVA assertions and 4 covergroups with a coverage-closure loop, reaching 100% functional coverage with 0 scoreboard errors and 0 assertion failures.
- Injected all 10 deliberate RTL bugs and confirmed the testbench caught each one.

UART Transmitter | Verilog HDL, Finite State Machines (FSM), RTL Design: GitHub

- Designed an FSM-based UART transmitter (IDLE/START/DATA/STOP) in Verilog with a shift register for serial data.
- Parameterized baud rate via CLKS_PER_BIT (100 MHz clock → 9600 baud).
- Verified using a self-checking testbench with a behavioral receiver model as monitor/checker, covering corner-case and random data (15 tests, all passing).
- Synthesized with Yosys to a gate-level netlist (171 cells, 34 flip-flops) and documented the synthesis results in the repo.

Blind Assistance Wearable Device | Arduino, C++, Ultrasonic Sensors: GitHub

- Designed a wearable obstacle and elevation-change detector on Arduino Uno using 2 ultrasonic sensors, alerting for obstacles within 80 cm and for raised surfaces or drops via downward-sensor thresholds.
- Implemented a priority-based alert logic with 5 distinct buzzer patterns, including a timeout fallback for sensor failures.
- Collaborated in a 3-member team on hardware integration, testing, and enclosure prototyping.

TECHNICAL SKILLS

Hardware Description Languages & Verification: Verilog HDL, SystemVerilog, SVA, constrained-random verification, functional coverage, self-checking testbenches
Programming and Scripting Languages: Python, C++, C, Tcl
Embedded Systems: Arduino, Microcontrollers, Sensor Integration
Tools & Platforms: Cadence Xcelium and Icarus Verilog (EDA Playground), Git, GitHub, Yosys, MATLAB, LTSpice.
Relevant Coursework & Topics: Digital Logic Design, Finite State Machines (FSMs), Computer Architecture Fundamentals, CPU Pipelines, Pipeline Hazards, Cache Memory, Microprocessor Architecture, Static Timing Analysis (STA), Data Structures and Algorithms (DSA)

INTERNSHIPS

Summer Intern – Bharat Heavy Electricals Limited (BHEL), Noida June 2026 – July 2026

- Studied thermal power plant operations and SCADA-based industrial control systems.
- Gained exposure to plant design tools within the Power Engineering Management department.

Summer Trainee/Intern – NTPC Limited, Noida June 2025 – July 2025

- Developed a Python/NLP PDF summarizer with Q&A, using document parsing and transformer-based models for automated summarization and natural-language querying of industrial reports.
- Gained exposure to data-centre infrastructure including servers, networking, cooling systems, and power management.

CERTIFICATIONS AND COURSES

- VLSI Design Flow: RTL to GDS - IIIT Delhi
- Verilog HDL: VLSI Hardware Design Comprehensive Masterclass on Udemy
- 170+ Questions solved on HDLBits.
- Data Structures & Algorithms in C++ – Coding Blocks
- McKinsey Forward Program – Leadership & Problem-Solving Training by McKinsey & Company

EXTRACURRICULAR ACTIVITIES

- Member, College Football Team.
- DELE A1 Spanish Certificate; German (beginner).